

Lecture-15

Mathematics 1 (15B11MA111)

Module: Triple Integrals

CO [C105.3]

Topics to be covered

➤ Triple Integrals

- ❖ Definition
- ❖ Examples
- ❖ Application of Triple Integral to Find Volume
- ❖ Examples
- ❖ Dirichlet's Integral
- ❖ Examples

➤ Reference for the lecture

R.K Jain and S.R.K. Iyenger, “Advanced Engineering Mathematics” fifth edition, Narosa publishing house, 2016.

Triple integrals

Let $f(x,y,z)$ be a continuous function defined over a closed and bounded region T in $\mathbb{R}^3$. Then the triple integral of $f(x,y,z)$ over T is given by

$$\iiint_T f(x, y, z) dx dy dz$$

Remark 1: Triple integral satisfies properties similar to double integral.

Remark 2: If $f(x,y,z) = 1$, then the triple integral

$$V = \iiint_T f(x, y, z) dx dy dz$$

gives the volume of the region T .

Evaluation of Triple Integrals:

We evaluate the triple integral by three successive integrations. If the region T can be described by

$$x_1 \leq x \leq x_2, y_1(x) \leq y \leq y_2(x), z_1(x, y) \leq z \leq z_2(x, y)$$

Then we evaluate the triple integral as

$$\begin{aligned} \int_{x_1}^{x_2} \int_{y_1(x)}^{y_2(x)} \int_{z_1(x,y)}^{z_2(x,y)} f(x, y, z) dz dy dx \\ = \int_{x_1}^{x_2} \left[\int_{y_1(x)}^{y_2(x)} \left[\int_{z_1(x,y)}^{z_2(x,y)} f(x, y, z) dz \right] dy \right] dx \end{aligned}$$

Example

Example 1: Evaluate the triple integral

$$\iiint_T y \, dx \, dy \, dz$$

Where T is the region bounded by the surfaces

$$x = y^2, x = y + 2, 4z = x^2 + y^2 \text{ and } z = y + 3$$

Solution:

Limits of z : $\frac{1}{4}(x^2 + y^2)$ to $y + 3$

Limits of x, y : we determine by the projection of T
in x-y plane i.e. the region bounded by the curves

$$x = y^2 \text{ and } x = y + 2.$$

These curves intersect at the points (1,-1) and (4, 2). Also

$$y^2 \leq y + 2 \text{ for } -1 \leq y \leq 2. \text{ Therefore,}$$

$$1 \leq y \leq 2, \quad y^2 \leq x \leq y + 2 \quad \text{and} \quad \frac{1}{4}(x^2 + y^2) \leq z \leq y + 3$$

Now we have,

$$\begin{aligned} \int_{-1}^2 \left[\int_{y^2}^{y+2} \left[\int_{\frac{1}{4}(x^2+y^2)}^{y+3} y dz \right] dx \right] &= \int_{-1}^2 \left[\int_{y^2}^{y+2} y \left\{ y + 3 - \frac{1}{4}(x^2 + y^2) \right\} dx \right] dy \\ &= \int_{-1}^2 \left[\left(y^2 + 3y - \frac{y^3}{4} \right) x - x^3 y / 12 \right]_{y^2}^{y+2} dy \\ &= \int_{-1}^2 \left[\left(y^2 + 3y - \frac{y^3}{4} \right) (y + 2 - y^2) - \frac{1}{12} y \{ (y + 2)^3 - y^6 \} \right] dy \end{aligned}$$

$$= \int_{-1}^2 \left[\frac{y^7}{12} + \frac{y^5}{4} - \frac{4y^4}{3} - 3y^3 + 4y^2 + \frac{16y}{3} \right] dy$$

$$= \left[\frac{y^8}{96} + \frac{y^6}{24} - \frac{4y^5}{15} - \frac{3y^4}{4} + \frac{4y^3}{3} + \frac{8y^2}{3} \right]_{-1}^2$$

$$= \frac{837}{160}$$

Application of Triple Integral to Find Volume

Example 1: Find the volume of the solid in the first octant bounded by the paraboloid

$$z = 36 - 4x^2 - 9y^2$$

Solution: We have

$$V = \iiint_T dz \, dy \, dx$$

The projection of the paraboloid (in the first octant) in the x-y plane is the region in the first quadrant of the ellipse

$$4x^2 + 9y^2 = 36 .$$

Therefore the region T is given by

$$0 \leq z \leq 36 - 4x^2 - 9y^2, 0 \leq y \leq \frac{1}{3}\sqrt{36 - 4x^2}, 0 \leq x \leq 3.$$

Hence,

$$V = \int_0^3 \left[\int_0^{\frac{2}{3}(\sqrt{9-x^2})} (36 - 4x^2 - 9y^2) dy \right] dx$$

$$= \int_0^3 [4(9 - x^2)y - 3y^3]_0^{\frac{2}{3}(\sqrt{9-x^2})} dx$$

$$= \frac{16}{9} \int_0^3 (9 - x^2)^{3/2} dx$$

Let us substitute $x = 3\sin\theta$

$$\begin{aligned} V &= \frac{16}{9} \int_0^{\pi/2} (27 \cos^3 \theta)(3 \cos \theta) d\theta \\ &= 27\pi \text{ cubic units} \end{aligned}$$

Example 2: Find the Volume of the solid enclosed between the surfaces

$$x^2 + y^2 = a^2 \quad \text{and} \quad x^2 + z^2 = a^2.$$

Solution: The limits of x , y , z are as:

$$-\sqrt{a^2 - x^2} \leq z \leq \sqrt{a^2 - x^2}, -\sqrt{a^2 - x^2} \leq y \leq \sqrt{a^2 - x^2}, -a \leq x \leq a.$$

Therefore,

$$V = \int_{-a}^a \int_{-\sqrt{a^2-x^2}}^{\sqrt{a^2-x^2}} \int_{-\sqrt{a^2-x^2}}^{\sqrt{a^2-x^2}} dz \, dy \, dx$$

$$V = \int_{-a}^a \left[\int_{-\sqrt{a^2-x^2}}^{\sqrt{a^2-x^2}} \left[\int_{-\sqrt{a^2-x^2}}^{\sqrt{a^2-x^2}} dz \right] dy \right] dx$$

$$V = 8 \int_0^a \int_0^{\sqrt{a^2-x^2}} \sqrt{a^2-x^2} \, dy \, dx$$

$$V = 8 \int_0^a (a^2 - x^2) \, dx = \frac{16}{3} a^3 \text{ cubic units}$$

Dirichlet's Integral

Let T be a closed region in the first octant in $\mathbb{R}^3$, bounded by the surface

$$(x/a)^p + (y/b)^q + (z/c)^r = 1 \quad \text{and the coordinate planes.}$$

Then, an integral of the form

$$I = \iiint_T x^{\alpha-1} y^{\beta-1} z^{\gamma-1} dx dy dz$$

is called a Dirichlet integral, where all the constants $\alpha, \beta, \gamma, a, b, c, p, q, r$ are assumed to be positive.

$$\iiint_T x^{\alpha-1} y^{\beta-1} z^{\gamma-1} dx dy dz = \frac{a^\alpha b^\beta c^\gamma}{pqr} \frac{\overline{\alpha/p} \overline{\beta/q} \overline{\gamma/r}}{\overline{\alpha/p + \beta/q + \gamma/r + 1}}$$

Examples

Example 1: Evaluate the Dirichlet integral

$$I = \iiint_T x^3 y^3 z^3 dx dy dz$$

Where T is the region in the first octant bounded by the sphere

$x^2 + y^2 + z^2 = 1$ and the coordinate planes.

Solution: Here $\alpha = \beta = \gamma = 4, a = b = c = 1, p = q = r = 2$

$$I = \frac{1}{8} \frac{\sqrt[2]{2} \sqrt[2]{2} \sqrt[2]{2}}{\sqrt[7]{7}} = \frac{1}{5760}$$

$$I = \iiint_T x^3 y^3 z^3 dx dy dz = \frac{1}{8}$$

Example 2: $I = \iiint_T x^{1/2} y^{1/2} z^{1/2} dx dy dz$

Where T is the region in the first octant bounded by the plane $x+y+z=1$ and the coordinate planes.

Solution: Here $\alpha = \beta = \gamma = \frac{3}{2}$, $a = b = c = 1$, $p = q = r = 1$

$$I = \iiint_T x^{1/2} y^{1/2} z^{1/2} dx dy dz$$

$$I = \frac{\overline{3/2} \overline{3/2} \overline{3/2}}{\overline{11/2}} = \frac{4\pi}{945}$$

Practice Problems

1. Find the volume of the solid which is bounded by the cylinder $x^2 + y^2 = 1$ and the planes $y+z=1$ and $z=0$.

Ans: π

2. Evaluate $I = \iiint dx dy dz$ where $(x/a)^2 + (y/b)^2 + (z/c)^2 = 1$

Ans: $\frac{1}{8}\pi abc$